

Unit 18: Shaped by Wind as a Geomorphic Agent

Reading: pages 260–273 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. Aeolian (wind) processes are most effective where surface material is
 - a) rich in clay and bound by calcium carbonate
 - b) frozen for most of the year
 - c) coarse, wet, and cemented together
 - d) fine, dry, and loose, with no vegetation cover
2. What share of Earth's deserts are partially or completely covered by sand?
 - a) About one-quarter
 - b) About one-half
 - c) About three-quarters
 - d) Nearly all of them
3. The cobblestone-like desert surface left behind after deflation removes the fine particles is called
 - a) a playa
 - b) desert pavement
 - c) a bajada
 - d) an erg
4. Barchan (crescent) dunes have horns that point
 - a) upwind
 - b) downwind
 - c) straight upward
 - d) in randomly varying directions
5. Saltation accounts for approximately what share of sand transport over dunes?
 - a) About 10%
 - b) About 20 to 25%
 - c) More than 50%
 - d) About 90%

6. A flat area of clay, silt, or sand encrusted with salt, left behind when a shallow desert lake dries up, is called
- a) a playa
 - b) an arroyo
 - c) a ventifact
 - d) a mesa

Part B · True or False

2 questions × 1 point = 2 points

7. Parabolic dunes form where sand accumulates in a moist environment with a cover of vegetation.
TRUE / FALSE
8. A mesa is defined as a flat-topped hill whose top is narrower than its height.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain the difference between *deflation* and *abrasion*, and name one landform produced by each.
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10. Explain how mesas and buttes are formed through *differential erosion*.
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